

Task 1

A) Data Clean

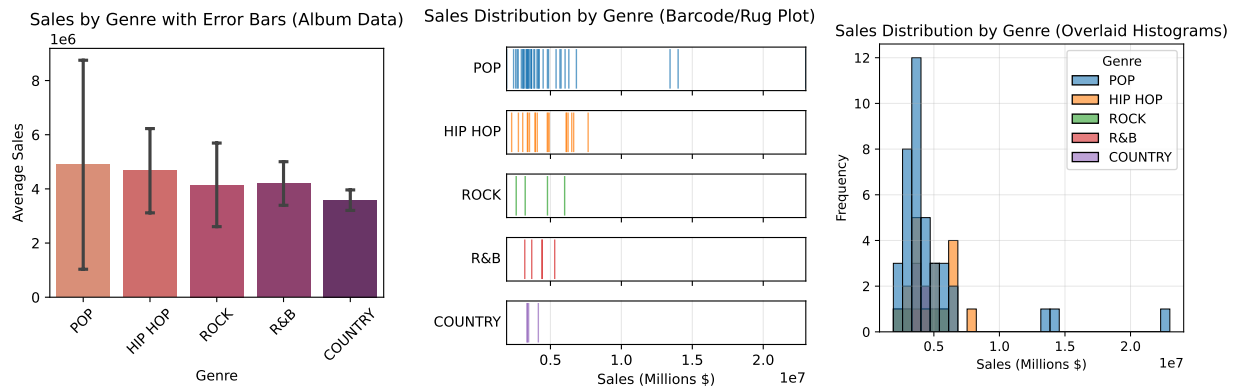
For cleaning the Album Data, we first need to filter any data entries that are not within the past decade. Then we go through the related columns and ensure all entries are consistent.

For cleaning the Anime Data, we first need to filter any data entries that are not within the past decade. Then we go through the related columns and ensure all entries are consistent.

For cleaning the algorithm data we need to ensure all entries are consistent with the parameters and the naming is consistent.

B) Generate Visualizations

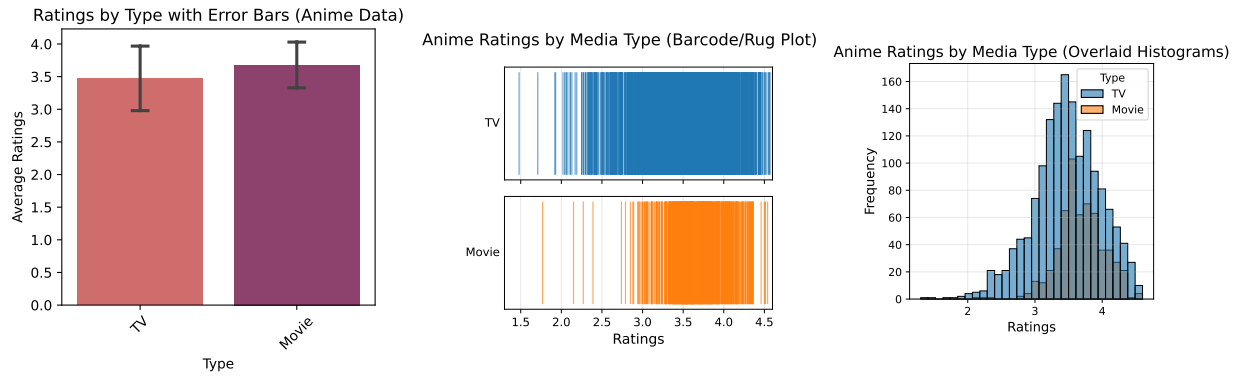
Album Data



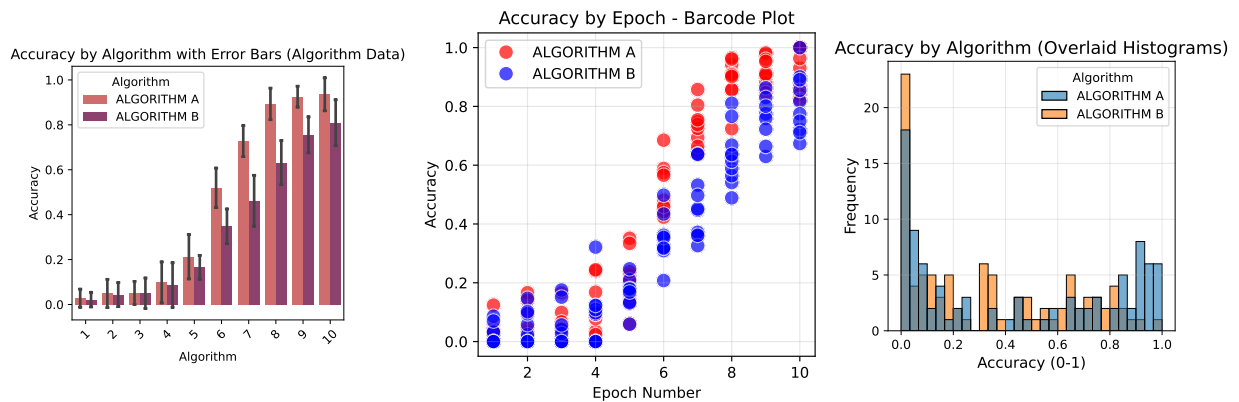
Above are the plots for the album data. The plots are Error Bar Plots, Barcode Plots, Histograms

Anime Data

Below are the plots for the anime data. The plots are Error Bar Plots, Barcode Plots, Histograms



Algorithm Data



Above are the plots for the algorithm data. The plots are Error Bar Plots, Barcode Plots, Histograms

C) Justify Best Visualization

Album Data

Error Bars

Pros:

- Nicely ordered by average sales
- Easy to look at

Cons:

- Pop has a large variance
- Population size is hidden
- Outliers hidden

Barcode Plot

Pros:

- Easy to see size of a category and the distribution / variance
- Outliers easily identifiable
- Not Cluttered

Cons:

- Hard to gauge the mean

Histogram

Pros:

- Can see the size of population / distribution / variance

Cons:

- Very cluttered
- Smaller categories are hard to see
- Hard to actually interpret any data

Best Visualization

The best visualization for the album data is the barcode plot because it is very decluttered (unlike the histogram) while still allowing to see accurate information about the overall distribution of the data, ie outliers and density. Additionally, despite the genres being vastly different in size we are still able to clearly see information on each genre accordingly.

Anime Data

Error Bar Plot

Pros:

- Easily identify mean and variance

Cons:

- Cannot get much insight on the data itself
- Unsure about population size / outliers

Barcode Plot

Pros:

- Can see that the TV population is much larger

Cons:

- Very cluttered and difficult to get any insight on the data

Histogram

Pros:

- Easily see average and overall distribution of each type
- Anime Types are slightly similar in size allowing to easily see both types
- Easy comparison between the two types

Cons:

- Slight size difference causes the movie type to always be behind the TV type

Best Visualization

Overall the best visualization for this dataset is definitely the histogram plot. This is because the graph is not cluttered and quite easy to read. Additionally unlike the error bar plot, the histogram allows us to look at the different sizes of the categories and compare the overall distribution.

Algorithm Data

Error Bar Plot

Pros:

- Easily compare change across the epochs
- Easily compare the two algorithms
- Easy to read

Cons:

- Early Epochs with low accuracy and high variance are slightly off

Barcode Plot

Pros:

- Can see the accuracy across the epochs
- In general can see that red performs better than blue

Cons:

- Data points tend to overlap causing clutter

Histogram

Pros:

- Easy to see how variant the data set is

Cons:

- No good way to plot to display the epoch number
- Quite clustered making it difficult to see

Best Visualization

Overall the best visualization is the error bar plot because it is very easy to analyze the differences across the different epoch while also comparing the two algorithms side by side. Additionally since the population sizes of each epoch are relatively small, the barcode plot tends to over due it while the histogram has no good way of representing the three categories (Algorithm, Accuracy and Epoch) because one of the Axis must be for frequency.

Task 2

A) Develop Hypothesis

Normal

For a normal distribution i have found a dataset for the height of a population. I hypothesize this data is normally distributed because it is highly clustered at an average point and very sparse at the extremes.

<https://www.kaggle.com/datasets/mustafaali96/weight-height?resource=download>

Uniform

I found a fake employee dataset and I hypothesize that the ID numbers of the employees are uniformly dsitributed because each ID should be unique to an individual.

<https://www.kaggle.com/datasets/ravindrasinghrana/employeeedataset>

Power Law Distribution

I choose population size of cities because there will be a higher frequency of lower population cities than higher population cities and the city size frequency does not decrease at an exponential rate.

<https://www.kaggle.com/datasets/dataanalyst001/population-of-all-us-cities-2024>

Exponential Distribution

For exponential distribution im hypothesizing that number of consecutive heads is exponentially distributed because an increase by one is approximately 50% decrease in probability.

<https://www.kaggle.com/datasets/irkanhadi/256-coin-flip-entries>

B) Fitting Distributions

Datasets		#Observations	Normal	Uniform
0	Height of People	10000	[66.37, 3.85]	[54.26, 79.0]
1	Employee Id Number	3000	[2500.5, 866.17]	[1001.0, 4000.0]
2	City Populations	300	[315577.41, 578032.61]	[110055.0, 8097282.0]
3	#Consecutive Heads	67	[2.01, 1.61]	[1.0, 11.0]

Datasets		#Observations	Power Law	Exponential
0	Height of People	10000	[8.65, 54.26]	0.08
1	Employee Id Number	3000	[4.8, 1001.0]	0.00
2	City Populations	300	[2.44, 110055.0]	0.00
3	#Consecutive Heads	67	[3.44, 1.0]	0.99

For estimating the normal and uniform parameters, they were obtained by finding the mean, std, min, and max of the datasets. Then the α was found using a powerlaw fit built into python,

this is a valid approach as it will find the most accurate α for each dataset. Similarly, λ was found using similar methods and is also valid with similar reasoning.

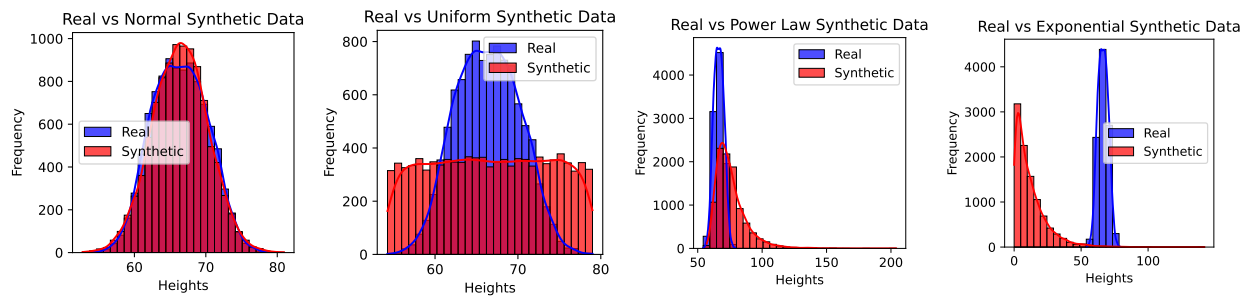
C) Comparing Real and Synthetic Data

Generating Synthetic Data

Here, for each dataset and each distribution type we generate a random population of datapoints given the information from the dataset following the specified distribution. Additionally, we ensure that this random population is of equal size to the original dataset for easier comparisons.

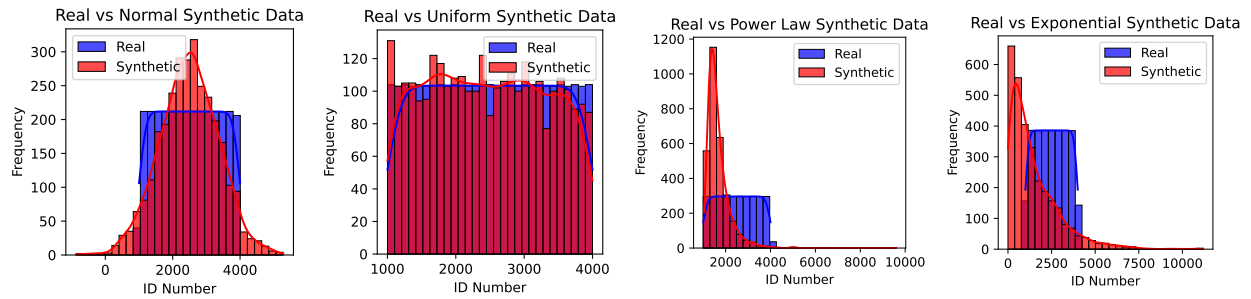
Generating Plots

Heights



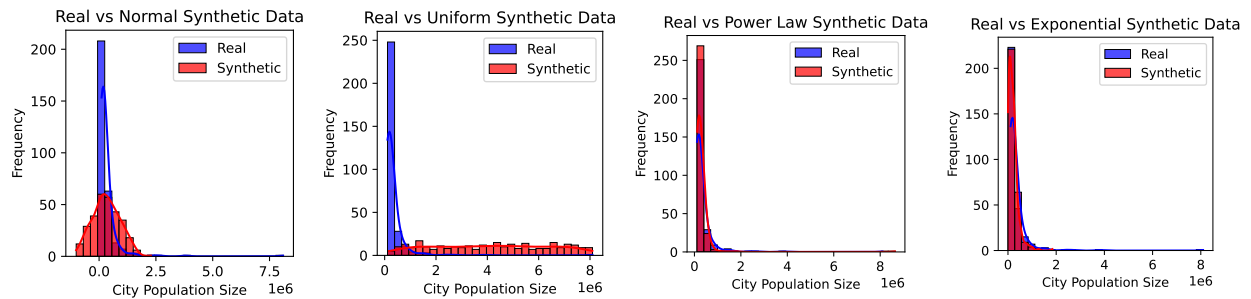
Above are the Real vs Synthetic plots for the height dataset. These include Normal, Uniform, Power Law, and Exponential Distributions.

Employee Ids



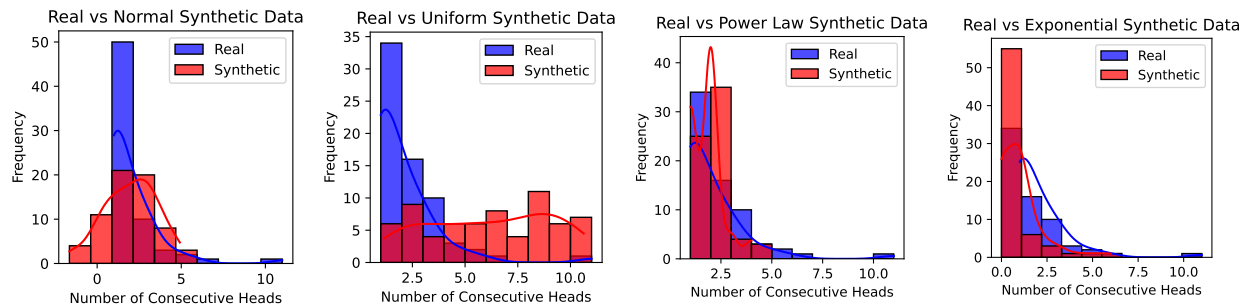
Above are the Real vs Synthetic plots for the Employee ID

City Populations



Above are the Real vs Synthetic plots for the City Populations dataset

Consecutive Heads



Above are the Real vs Synthetic plots for the Consecutive Heads dataset

Evaluating Initial Hypothesis

Heights

For the Heights Dataset, the synthetic dataset that is most similar to the model is the Normal dataset. This means that the Heights Dataset behave as expected and another model was not a better fit.

Employee ID

For the Employee ID Dataset, the synthetic dataset that is most similar to the model is the Uniform dataset. This means that the Employee ID Dataset behave as expected and another model was not a better fit.

City Population Sizes

For the City Population Sizes dataset, the Power Law and the Exponential synthetic datasets are the ones that most match the original. Personally, I think the power law synthetic data matches slightly more than the exponential law. This means that the dataset does behave as expected however there is a potential argument for the exponential model to be a better fit.

Consecutive Heads

For the consecutive heads data, the synthetic dataset that is the most similar to the original is the exponential dataset. Thus the dataset behaved as expected and there is not another model that would be a better fit.